MySQL Assignment 2 – Querying Data

Objective:

The objective of this assignment is to perform data retrieval and analysis using MySQL query commands on the Employee Database created in Assignment 1. This assignment aims to demonstrate the use of SQL statements such as SELECT, DISTINCT, WHERE, ORDER BY, GROUP BY, HAVING, aggregate functions, aliases, and different types of joins to extract meaningful information from the database. It also focuses on updating existing data and generating reports by querying employee, department, and location details efficiently.

DML insert codes

Inserting for Department Table:

```sql
-- Assignment 2 - Querying Data
USE EmployeeDB;
```

```sql
INSERT INTO Departments (department_id, department_name) VALUES
(1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
(5, 'Product Management'),
(6, 'Content Creation'),
(7, 'Finance'),
(8, 'Design'),
(9, 'Research and Development'),
(10, 'Customer Support'),
(11, 'Business Development'),
(12, 'IT'),
(13, 'Operations');
```

48	11:17:37	USE EmployeeDB		0 row(s) affected

```sql
SELECT * FROM Departments;
```

Result Grid | Filter Rows: | Edit:

department_id	department_name
11	Business Development
6	Content Creation
10	Customer Support
3	Data Science
8	Design
7	Finance
4	Human Resources
12	IT
2	Marketing
13	Operations
5	Product Management
9	Research and Develo...
1	Software Development
NULL	NULL

49	11:19:20	INSERT INTO Departments (department_id, department_name) VALUES (1, 'Software Development'), (2, 'Marketing'), (3, 'Data Science'), (4, 'Human R...	13 row(s) affected Records: 13 Duplicates: 0 Warnings: 0
50	11:20:03	SELECT * FROM Departments LIMIT 0, 1000	13 row(s) returned

Inserting for Location Table:

```sql
INSERT INTO Locations (location_name) VALUES
('Chennai'),
('Bangalore'),
('Hyderabad'),
('Pune');
```

```sql
SELECT * FROM Locations;
```

Result Grid | Filter Rows: | Edit:

location_id	location_name
2	Bangalore
1	Chennai
3	Hyderabad
4	Pune
NULL	NULL

51	11:24:15	INSERT INTO Locations (location_name) VALUES ('Chennai'), ('Bangalore'), ('Hyderabad'), ('Pune')	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
52	11:26:00	SELECT * FROM Locations LIMIT 0, 1000	4 row(s) returned

Inserting for Employees Table:

```sql
INSERT INTO Employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, locati
(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
(5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),
(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),
(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),
(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),
(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),
(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),
(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),
(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),
(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),
(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),
(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),
(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),
(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),
(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),
(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),
(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),
(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),
(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),
(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),
(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),
(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),
(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),
(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),
(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),
(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),
(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);
```

```sql
SELECT * FROM Employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	3	4
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	80000.00	12	1
5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	45000.00	10	2
5004	Kiara Malhotra	F	29	2015-07-05	NULL	70000.00	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	55000.00	11	1
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	65000.00	8	2
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	2	3
5008	Diya Jha	F	27	2016-03-05	Graphic Designer	70000.00	8	4
5009	Kiaan Desai	M	30	2016-05-20	Sales Executive	55000.00	11	3
5010	Atharv Yadav	M	29	2016-07-10	Systems Administrator	80000.00	12	4
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	60000.00	2	1
5012	Myra Verma	F	26	2016-11-05	Operations Manager	95000.00	13	2
5013	Arnav Rao	M	33	2017-01-20	Customer Success Manager	75000.00	10	3
5014	Vihaan Mohan	M	30	2017-03-10	Supply Chain Analyst	60000.00	10	2
5015	Ishaan Kumar	M	27	2017-05-20	Financial Analyst	85000.00	7	1
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	100000.00	4	4
5017	Kabir Nair	M	28	2017-09-11	IT Support Specialist	80000.00	12	2
5018	Ishan Mishra	M	25	2017-11-20	Research Scientist	75000.00	9	3
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	55000.00	4	4
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	85000.00	7	3
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	4	4
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	95000.00	5	1

53	11:27:34	INSERT INTO Employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary) VALUES (5001, 'Viha...	30 row(s) affected Records: 30 Duplicates: 0 Warnings: 0
54	11:28:43	SELECT * FROM Employees LIMIT 0, 1000	30 row(s) returned

1. Distinct Values:

◉ a query to retrieve distinct salaries from the Employees table.

```sql
SELECT DISTINCT Salary FROM Employees;
```

Salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

55 11:33:41 SELECT DISTINCT Salary FROM Employees LIMIT 0, 1000 — 11 row(s) returned

2. Alias (AS):

◉ Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```sql
SELECT age AS Employee_Age,
Salary AS Employee_Salary
FROM Employees;
```

Employee_Age	Employee_Salary
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00
29	80000.00
28	60000.00
26	95000.00
33	75000.00
30	60000.00
27	85000.00

59 11:38:32 SELECT age AS Employee_Age, Salary AS Employee_Salary FROM Employees LIMIT 0, 1000 — 30 row(s) returned

3. Where Clause & Operators:

◉ Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```sql
SELECT * FROM Employees
WHERE Salary > 50000
AND hire_date < '2016-01-01';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	3	4
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	80000.00	12	1
5004	Kiara Malhotra	F	29	2015-07-05	NULL	70000.00	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	55000.00	11	1
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	65000.00	8	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3	11:19:25	SELECT * FROM Employees WHERE Salary > 50000 AND hire_date < '2016-01-01' LIMIT 0, 1000	5 row(s) returned

◉ Find the employee whose designation is missing and fill it with "Data Scientist".

```sql
SELECT * FROM Employees
WHERE Designation IS NULL;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id
5004	Kiara Malhotra	F	29	2015-07-05	NULL	70000.00	8	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```sql
SET SQL_SAFE_UPDATES = 0;
UPDATE Employees
SET designation='Data Scientist'
WHERE designation IS NULL;
SET SQL_SAFE_UPDATES = 1;
SELECT * FROM Employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	80000.00	12	1
5003	Aaradhya Iyer	F	26	2015-05-20	Customer Support Executive	45000.00	10	2
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	70000.00	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	55000.00	11	1
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	65000.00	8	2
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	2	3

#	Time	Action	Message
8	11:29:01	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
9	11:29:01	UPDATE Employees SET designation='Data Scientist' WHERE designation IS NULL	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
10	11:30:08	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
11	11:30:14	UPDATE Employees SET designation='Data Scientist' WHERE designation IS NULL	0 row(s) affected Rows matched: 0 Changed: 0 Warnings: 0
12	11:32:42	SET SQL_SAFE_UPDATES = 1	0 row(s) affected
13	11:33:05	SELECT * FROM Employees LIMIT 0, 1000	30 row(s) returned

Sorting and Grouping Data:

1. ORDER BY:

◉ Find employees sorted by department ID in ascending order and salary in descending order.

```sql
SELECT * FROM Employees
ORDER BY department_id ASC,
salary DESC;
```

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	2	3
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	60000.00	2	1
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	100000.00	3	4
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	75000.00	3	2
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	3	4
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	100000.00	4	4
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	4	4
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	55000.00	4	4
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	95000.00	5	1
5030	Sara Iyer	F	28	2019-11-20	Project Manager	90000.00	5	1
5027	Ananya Paul	F	32	2019-05-20	Content Writer	60000.00	6	3
5028	Anaya Kapoor	F	26	2019-07-05	Event Coordinator	60000.00	6	1
5015	Ishaan Kumar	M	27	2017-05-20	Financial Analyst	85000.00	7	1
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	85000.00	7	3
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	70000.00	8	3
5008	Diya Jha	F	27	2016-03-05	Graphic Designer	70000.00	8	4
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	65000.00	8	2

Action Output

#	Time	Action	Message
1	12:32:54	SELECT * FROM Employees ORDER BY department_id ASC, salary DESC LIMIT 0, 1000	30 row(s) returned

2. LIMIT:

◉ Display the first 5 employees hired in the year 2018.

```sql
SELECT * FROM Employees
WHERE YEAR(hire_date) = 2018
LIMIT 5;
```

employee_id	Employee_name	Gender	Age	Hire_date	Designation	Salary	department_id	location_id
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	55000.00	4	4
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	85000.00	7	3
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	4	4
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

#	Time	Action	Message
2	12:40:24	SELECT * FROM Employees WHERE YEAR(hire_date) = 2018 LIMIT 5	5 row(s) returned

3. Aggregate Functions:

◉ Calculate the sum of all salaries in the Finance department.

```sql
SELECT SUM(e.salary) AS Total_Salary
FROM Employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name = 'Finance';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Salary
170000.00

3 12:55:42 SELECT SUM(e.salary) AS Total_Salary FROM Employees e JOIN departments d ON e.department_id = d.department_id WHERE d.department_name = ... 1 row(s) returned

◉ Find the minimum age among all employees.

```sql
SELECT MIN(age) AS Minimum_Age
FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Minimum_Age
25

5 12:58:30 SELECT MIN(age) AS Minimum_Age FROM Employees LIMIT 0, 1000 1 row(s) returned

4. GROUP BY:

◉ List the maximum salary for each location.

```sql
SELECT location_id,
MAX(salary) AS Maximum_Salary
FROM Employees
GROUP BY location_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	Maximum_Salary
1	95000.00
2	95000.00
3	90000.00
4	100000.00

9 13:03:01 SELECT location_id, MAX(salary) AS Maximum_Salary FROM Employees GROUP BY location_id LIMIT 0, 1000 4 row(s) returned

◉ Calculate the average salary for each designation containing the word 'Analyst'.

```sql
SELECT designation,
AVG(salary) AS Average_Salary
FROM Employees
WHERE designation LIKE '%Analyst%'
GROUP BY designation;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conter

designation	Average_Salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

10 13:06:03 SELECT designation, AVG(salary) AS Average_Salary FROM Employees WHERE designation LIKE '%Analyst%' GROUP BY designation LIMIT 0, 1000 7 row(s) returned

5. HAVING:

◉ Find departments with less than 3 employees.

```sql
-- Departments having less than 3 Employees
SELECT department_id,
COUNT(*) AS Total_Employees
FROM Employees
GROUP BY department_id
HAVING COUNT(*)<3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	Total_Employees
1	1
5	2
6	2
7	2
9	1
11	2
13	1

12 13:13:09 SELECT department_id, COUNT(*) AS Total_Employees FROM Employees GROUP BY department_id HAVING COUNT(*)<3 LIMIT 0, 1000 7 row(s) returned

◉ **Find locations with female employees whose average age is below 30.**

```sql
-- Locations with female employees whose average age is below 30
SELECT location_id,
ROUND(AVG(age), 2) AS Average_Age
FROM Employees
WHERE gender = 'F'
GROUP BY location_id
HAVING AVG(age)<30;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	Average_Age
1	26.75
2	27.33
4	29.20

14 13:20:41 SELECT location_id, ROUND(AVG(age), 2) AS Average_Age FROM Employees WHERE gender = 'F' GROUP BY location_id HAVING AVG(age)<30 ... 3 row(s) returned

Joins:

1. Inner Join:

◉ **List employee names, their designations, and department names where employees are assigned to a department.**

```sql
-- Inner Join
SELECT e.employee_name,
e.designation,
d.department_name
FROM Employees e
INNER JOIN Departments d
ON e. department_id = d. department_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	designation	department_name
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science

15 22:16:34 SELECT e.employee_name, e.designation, d.department_name FROM Employees e INNER JOIN Departments d ON e. department_id = d. department... 30 row(s) returned

2. Left Join:

◉ List all departments along with the total number of employees in each department, including departments with no employees.

```
SELECT d.department_name,
COUNT(e.employee_id) AS Total_Employees
FROM Departments d
LEFT JOIN Employees e
ON d.department_id = e.department_id
GROUP BY
d.department_id,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_name	Total_Employees
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3
Operations	1
Product Management	2
Research and Develo...	1
Software Development	1

16 22:26:44 SELECT d.department_name, COUNT(e.employee_id) AS Total_Employees FROM Departments d LEFT JOIN Employees e ON d.department_id = e.de... 13 row(s) returned

3. Right Join:

◉ Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
-- Right Join
SELECT
l.location_name,
e.employee_name
FROM Employees e
RIGHT JOIN Locations l
ON e.location_id=l.location_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_name	employee_name
Bangalore	Aaradhya Iyer
Bangalore	Dhruv Shetty
Bangalore	Myra Verma
Bangalore	Vihaan Mohan
Bangalore	Kabir Nair
Bangalore	Anika Paul
Bangalore	Vivaan Singh
Bangalore	Arjun Kumar
Chennai	Reyansh Singh
Chennai	Anvi Chaudhary
Chennai	Saanvi Patel
Chennai	Ishaan Kumar
Chennai	Aarav Nair
Chennai	Aryan Shetty
Chennai	Anaya Kapoor

17 22:31:07 SELECT l.location_name, e.employee_name FROM Employees e RIGHT JOIN Locations l ON e.location_id=l.location_id LIMIT 0, 1000 30 row(s) returned

Conclusion:

This assignment successfully demonstrated the use of MySQL querying techniques to retrieve, filter, sort, group, and analyze employee data from the database. Various SQL commands, including DISTINCT, WHERE, ORDER BY, GROUP BY, HAVING, aggregate functions, and joins, were executed to generate meaningful insights from the employee, department, and location tables. The assignment enhanced the understanding of relational database querying, data manipulation, and table relationships, while reinforcing the practical application of SQL for efficient data analysis and reporting.